

UTKARSH BARNWAL

Greater Noida,Uttar Pradesh

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EDUCATION

Lloyd Institute of Engineering and Technology

Bachelor of Technology (B.Tech) CGPA: 7.8

Jun 2023 – Jun 2026

Greater Noida, Uttar Pradesh

TECHNICAL SKILLS

- Languages:** C++, JavaScript, TypeScript, Python
Frontend: React.js, Next.js, Tailwind CSS, Redux, Zustand
Backend: Node.js, Express.js, REST APIs, JWT Authentication
Databases: MongoDB, Redis, SQL, Pinecone (Vector DB)
DevOps & Tools: Docker, Nginx, Git, GitHub, AWS (Basics)
AI & Systems: RAG, LLM APIs (OpenAI, Gemini), Custom AI Agents, Microservices, RabbitMQ

PROJECTS

Aurora – LLM Powered RAG System | MERN, RAG, Gemini API, Pinecone

June 2025

- Developed a full-stack **LLM-powered assistant** using Retrieval-Augmented Generation (RAG) for contextual responses over custom datasets.
- Engineered **vector embeddings and semantic search** using Pinecone to enhance retrieval accuracy and reduce hallucinations.
- Built scalable backend services with **Node.js and Express** to handle AI inference and retrieval workflows efficiently.
- Integrated secure **JWT-based authentication** and session management for personalized AI interactions.

E-Commerce Backend with AI Agent | Node.js, Express, MongoDB, Redis, Docker

July 2025

- Architected a **microservices-based backend** managing products, cart, orders, authentication, and payments.
- Developed 20+ RESTful APIs with optimized MongoDB schemas for efficient querying and data consistency.
- Improved performance using **Redis caching**, reducing redundant database calls and response latency.
- Created an **AI automation agent** capable of simulating user actions such as cart management and session workflows.
- Containerized services using Docker and enforced **role-based access control (RBAC)** with JWT authentication.

HACKATHONS

Reimagine Hackathon

2025

Frontend Hackathon by Sheryians

- Developed responsive UI clones using React and Tailwind CSS with reusable component architecture.
- Optimized layouts for accessibility, responsiveness, and performance improvements.

Phantom Agents Hackathon

2025

AI Hackathon

- Designed modular AI agents for autonomous workflow execution using LangChain and LLM APIs.
- Integrated vector databases (Pinecone/Redis) to enable context-aware multi-step reasoning.
- Architected scalable multi-agent workflows ensuring efficient inter-agent communication.

CERTIFICATIONS

- .75● Frontend Development (React, Tailwind CSS) – Sheryians
- .75● Backend Development (Node.js, Express.js, MongoDB) – Sheryians
- .75● Projects & Hackathons – Sheryians